

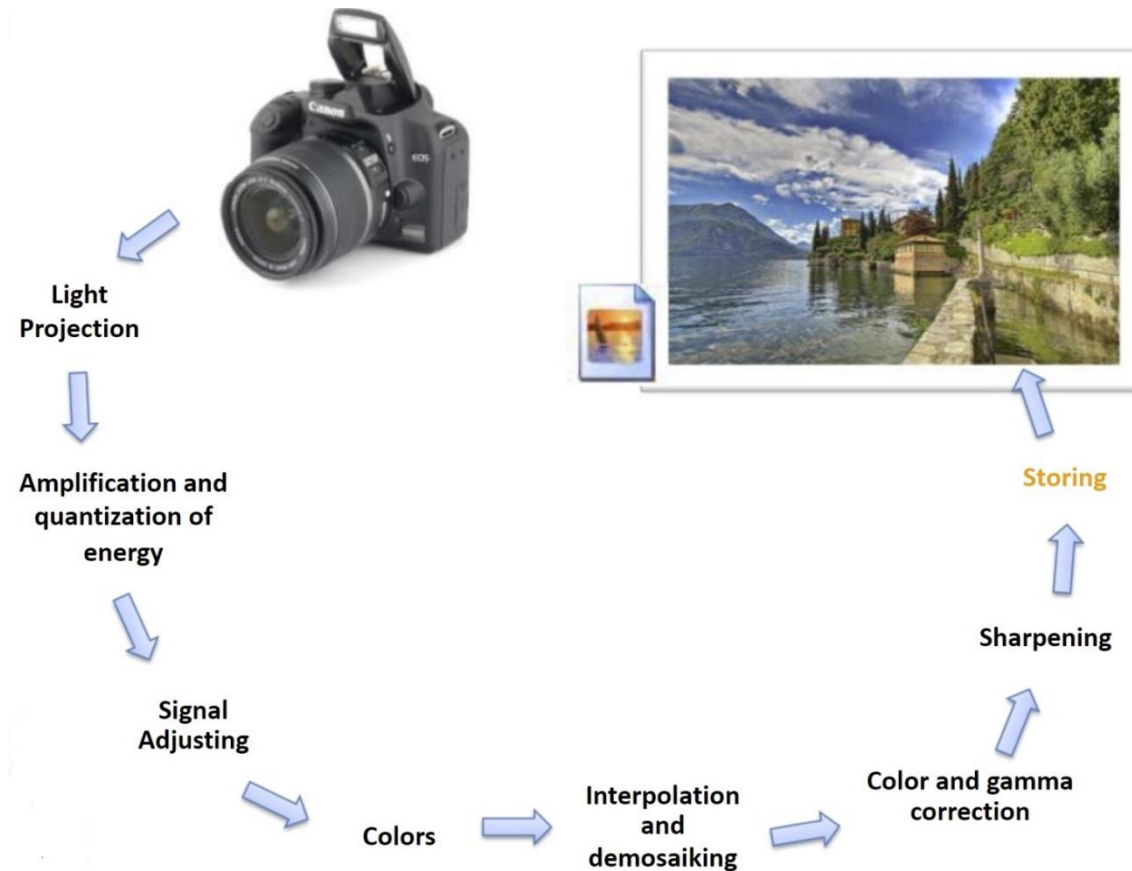


# Image Editing Forensics: inconsistency in lighting

Taras Holotyak

# Introduction

## Visual information processing chain



# Introduction

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Possible applications of digital image forensics to:

- Traces left in image acquisition (PRNU patterns)
- Traces left in image storage (JPEG compression)
- Traces left in image editing
  - Inconsistency in lighting
  - Local filtering traces
  - Detection of copy-move attacks
  - Resampling detection
  - Blind image-splicing detection

# Traces of image editing

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## 1. Inconsistency in lighting

*Copy-paste image tampering is the most common method of creating an image forgery. Usually, they are causing some inconsistencies in lighting, shadows, perspective, etc.*

*If two objects originally come from different images, it is unlikely that their light sources will be similar in direction and distance.*

# Traces of image editing

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## 1. Inconsistency in lighting

- Lighting traces

*When an image is captured, the objects captured in the scene are illuminated by a light source. This light source will leave traces on the objects, such as intensity differences on the surface, and shadows. Hence, the light direction can be inferred from these traces. If the objects in an image are illuminated by different light sources, these objects are very unlikely to belong to the same original scene, which indicates forgery.*

### Fonda Speaks To Vietnam Veterans At Anti-War Rally



Actress And Anti-War Activist Jane Fonda Speaks to a crowd of Vietnam Veterans as Activist and former Vietnam Vet John Kerry (LEFT) listens and prepares to speak next concerning the war in Vietnam (AP Photo)



# Traces of image editing

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## 1. Inconsistency in lighting

- Image-tampering detection using a lighting pattern

*The lighting environment in the real world is complex. Light sources are three dimensional (3D), and sometimes there are multiple light sources. Hence, assumptions are made to simplify the problem: the Lambertian assumption of the surface of interest, the constant assumption of the object's reflectance, and the assumption that the light source is located infinitely far away.*



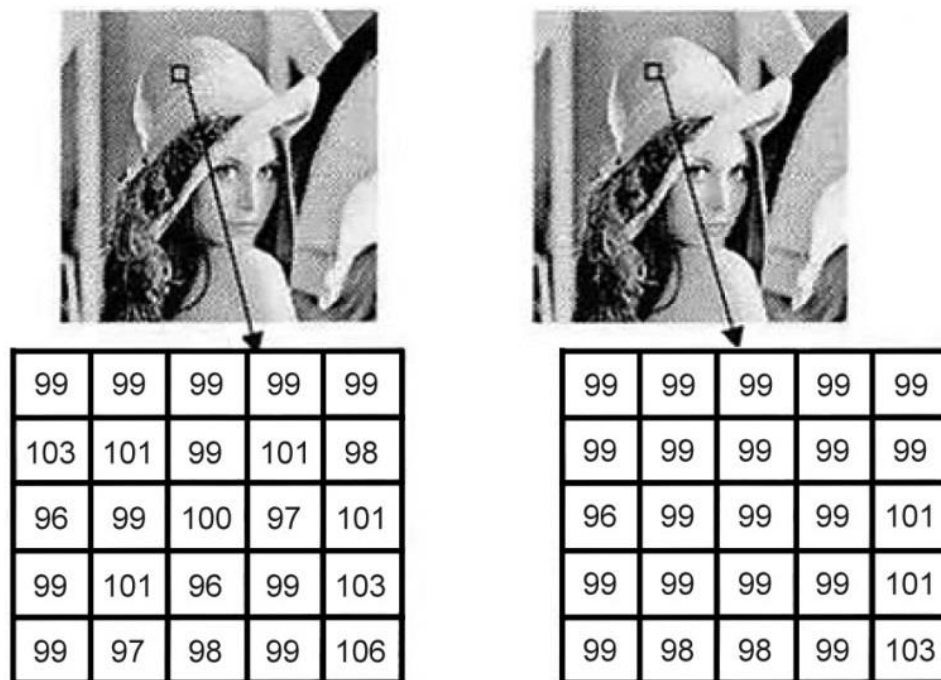
# Traces of image editing

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## 2. Local filtering traces

- Median filtering detection

*Median filtering is a common image post-processing method to filter out image noise. With its nonlinear property, it also can be used in image forgeries to remove specific modification traces.*

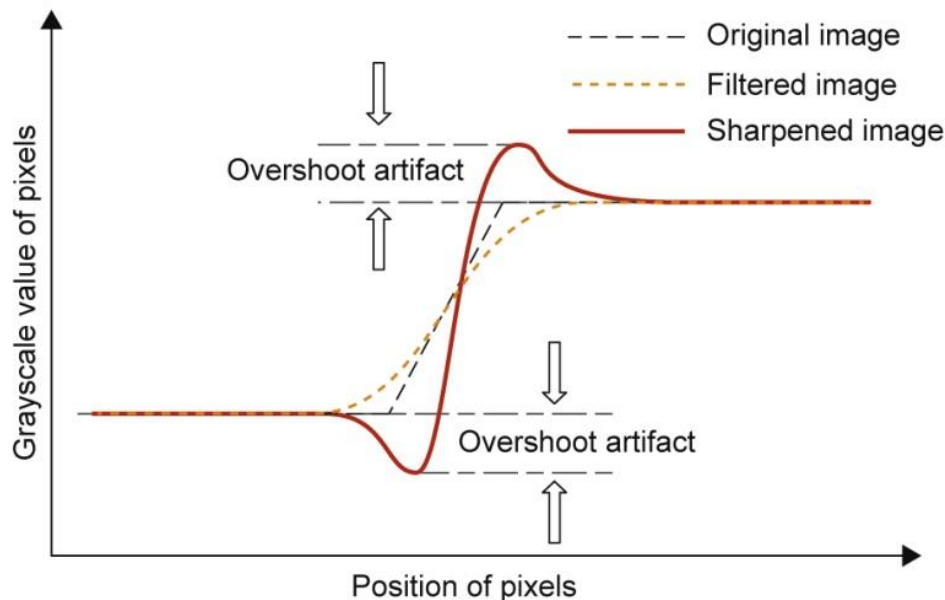


# Traces of image editing

## 2. Local filtering traces

- Unsharp masking sharpening detection

*Unsharp masking sharpening is a technique that is widely used to enhance edge contrast in order to improve image quality. However, in many image forgeries, Unsharp masking sharpening can also be employed to cover the traces of image manipulation.*





# Traces of image editing

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## 3. Detection of copy-move attacks

*The aim of a copy-move attack is to deliberately forge or remove one or more objects in the source image. For **object creation**, the new object is generated by copying a source object and moving it to another position in the same image. For **object removal**, the object region being removed is replaced by some background patches of the same image. In order to deceive the human eye, fundamental affine transforms such as scaling, rotation are usually applied before pasting.*

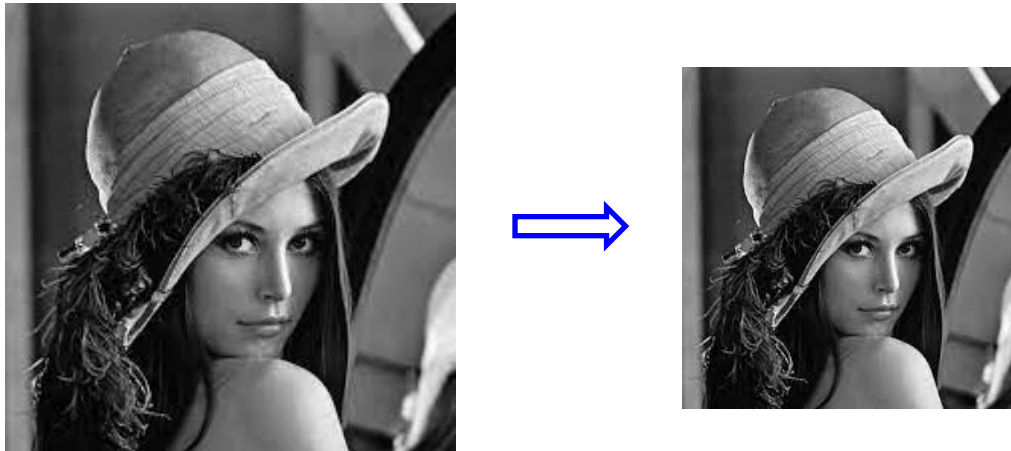


# Traces of image editing

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## 4. Resampling detection

*The splicing image region usually undergoes a resizing and/or a rotation procedure to cover the forgery traces and to make the final image forgery appear more realistic. Moreover, in the resizing and rotation procedure, the pixels in the modified image patch must be resampled to fit the new sample lattice. Hence, resampling detection can help to detect possible image forgeries and locate suspicious splicing regions.*



# Traces of image editing

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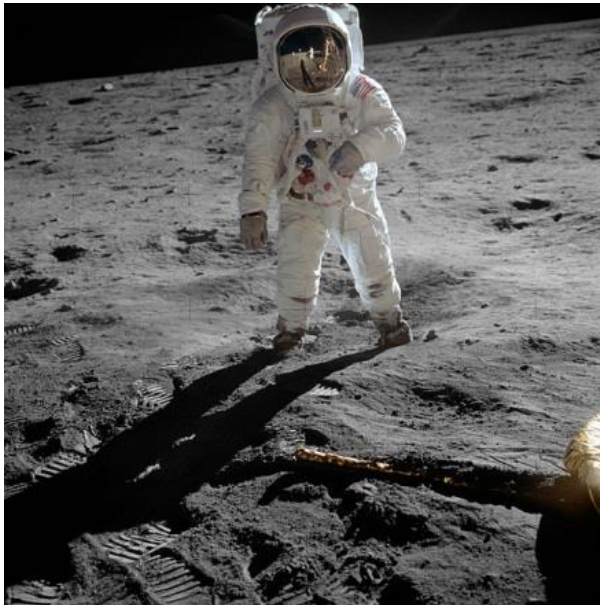
## 5. Blind image-splicing detection

*This kind of method attempts to detect various image-tampering and manipulation procedures based on the fact that such procedures will inevitably introduce certain artifacts that do not exist in the original images. These approaches lead to a binary decision regarding whether a test image is original or not.*



# Photo manipulation with inconsistent shadows

## Motivation



© NASA (<http://www.nasa.gov/>)



© TIME Inc. (<http://www.time.com/>)

# Photo manipulation with inconsistent shadows

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## Motivation

- 3-D lighting effects and shadows can be difficult to forge
- Lighting effects and shadows survive common operations
- In complex settings:
  - Visual system is less capable at detecting inconsistencies
  - Variety of shadows, shapes and surfaces

## Objective

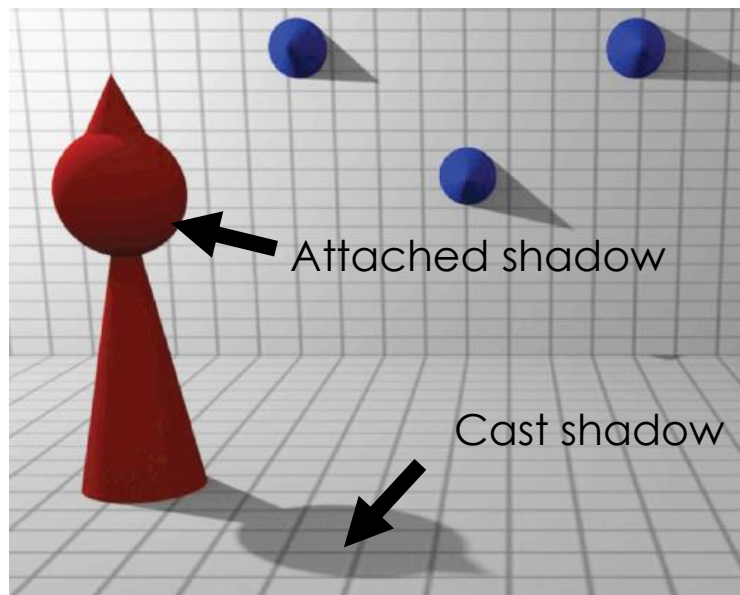
- Detect inconsistent shadows in images
- Support the decision:
  - Is a given photograph fake or not?

# Photo manipulation with inconsistent shadows

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## Basic assumptions

- Two types of shadows can be analyzed:
  - Cast Shadows
  - Attached Shadows
- A single distance or light source
- No other assumptions on scene geometry





# Photo manipulation with inconsistent shadows

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## Example



# Photo manipulation with inconsistent shadows

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## Example





# Photo manipulation with inconsistent shadows

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## Example



# Photo manipulation with inconsistent shadows

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## Methodology

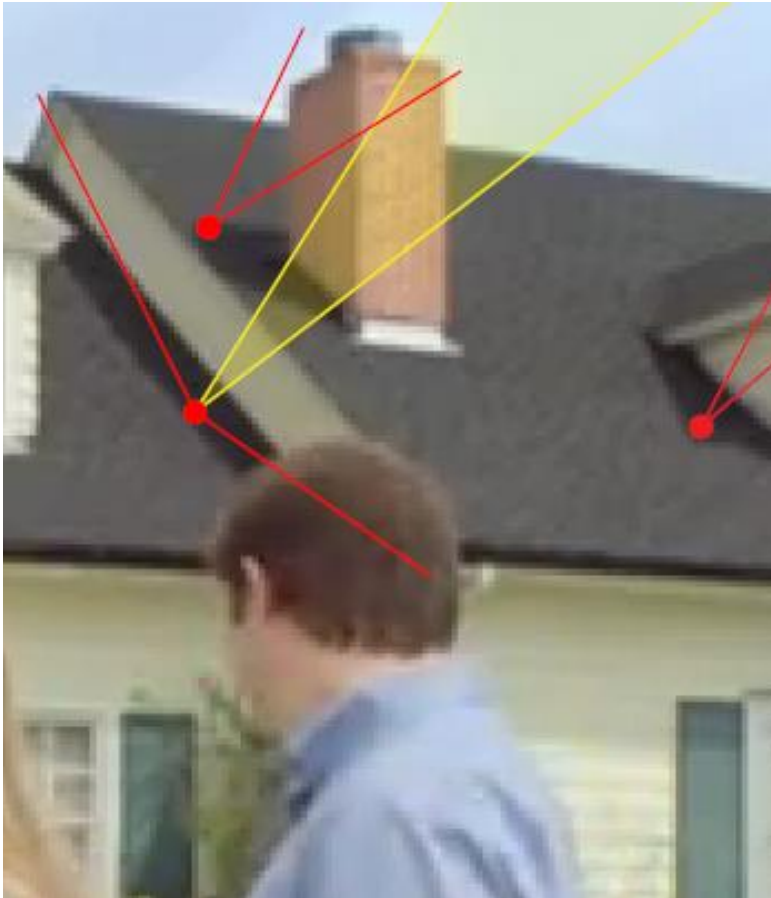


Wedge-shaped constraints describe direction from a point in a cast shadow to points on an object that may have cast the shadow.

# Photo manipulation with inconsistent shadows

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## Methodology



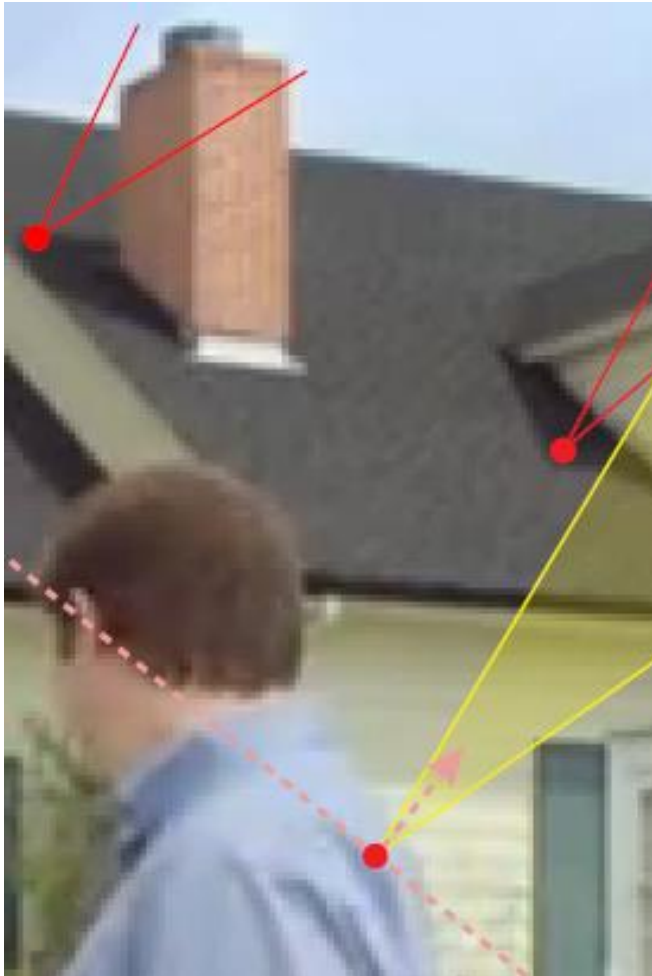
Range of plausible light directions automatically rendered (yellow wedge)

- Constraint ignored if yellow wedge is too narrow to encompass the object which cast the shadow

# Photo manipulation with inconsistent shadows

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## Methodology



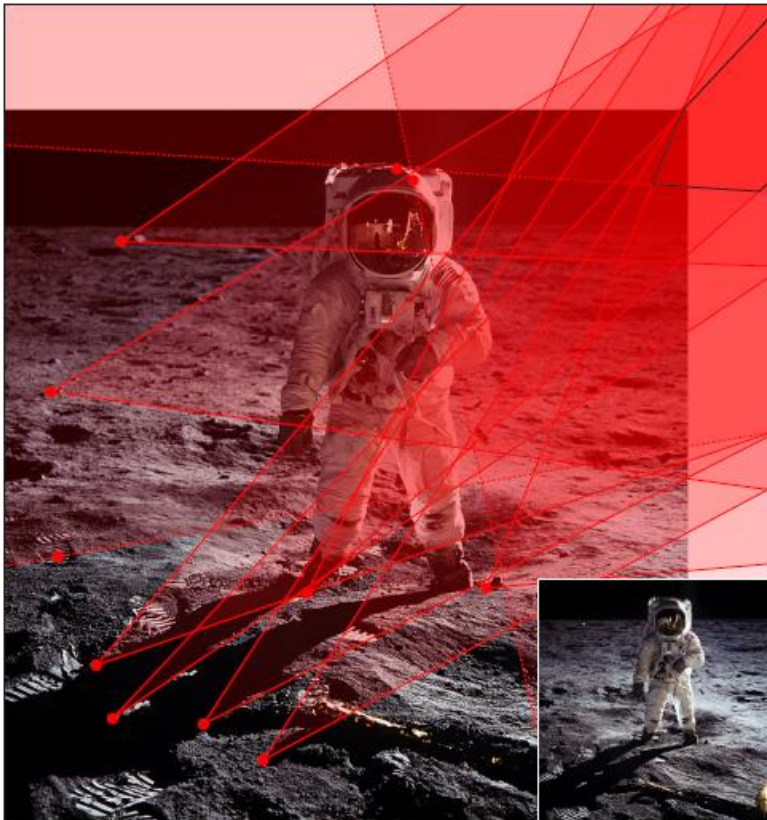
**Attached shadows** are specified with a half-plane

- Orientation and direction automatically suggested by computing image gradient
- Analyst can accept or adjust the constraint

# Photo manipulation with inconsistent shadows

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## Methodology



Constrain the projection location of the light source

- Feasible solutions indicate physically plausible shadows
- Failure to find a solution provides evidence of photo tampering

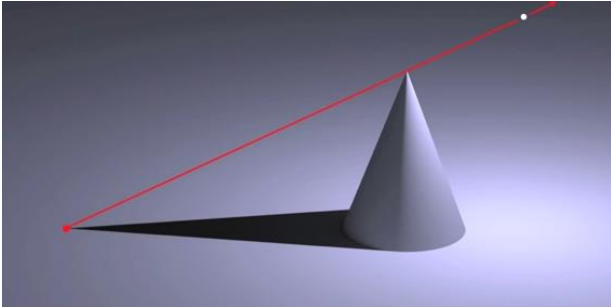
Forensic analyst needs to specify shadow's constraints

If a wedge does not overlap the yellow wedge at all, the constraint is inconsistent with previous ones: forgery

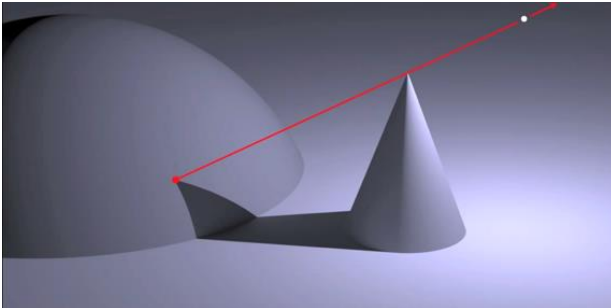
# Photo manipulation with inconsistent shadows

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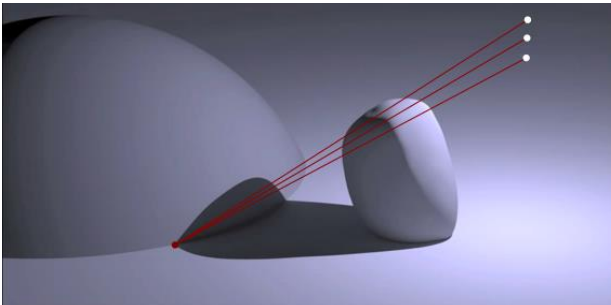
## Cast shadow



Where is the light source?  
Easy case



Easy case



Difficult case

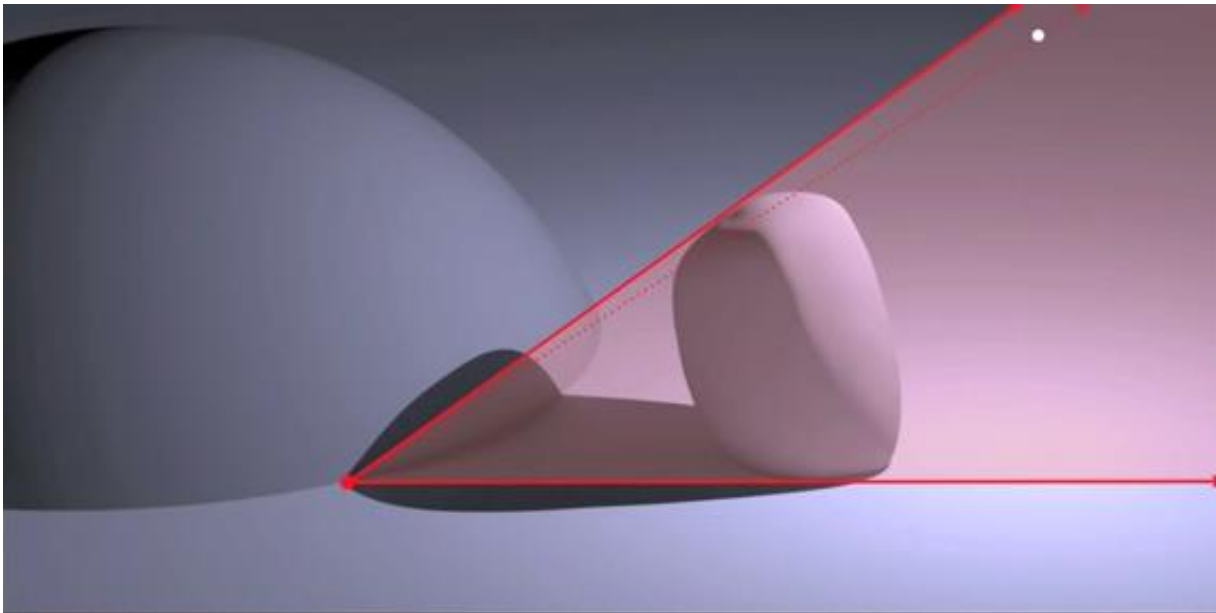


# Photo manipulation with inconsistent shadows

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## Cast shadow

Solution: constrain the position to a region inside a wedge

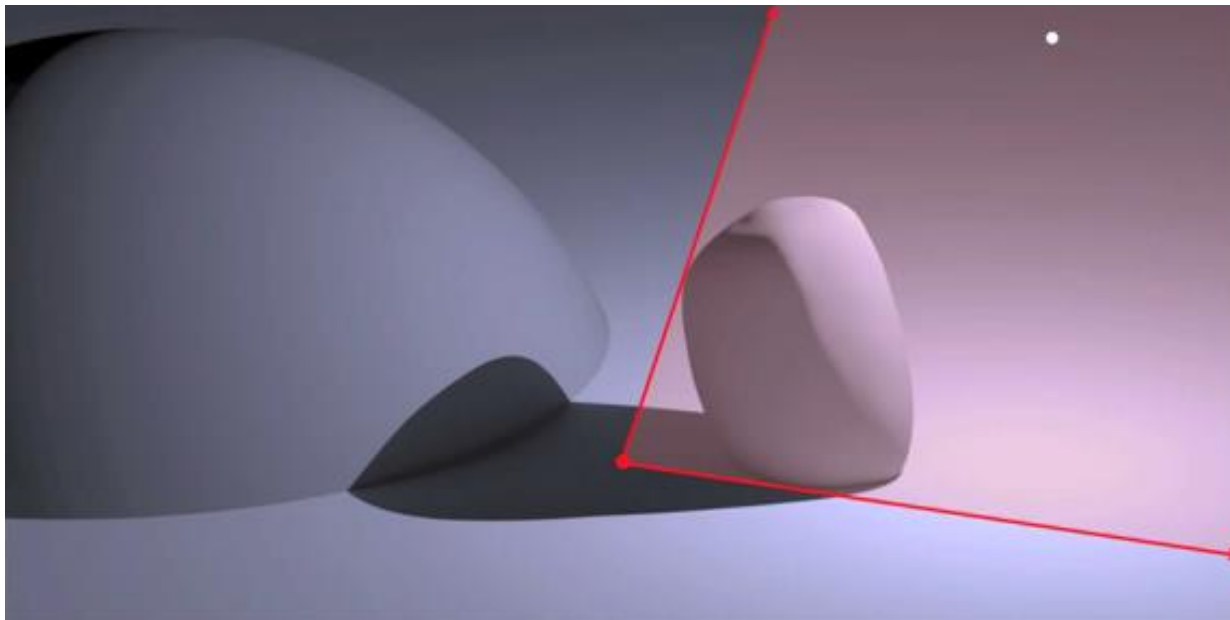


# Photo manipulation with inconsistent shadows

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## Cast shadow

Due to such solution, now we can choose any point inside the shadow

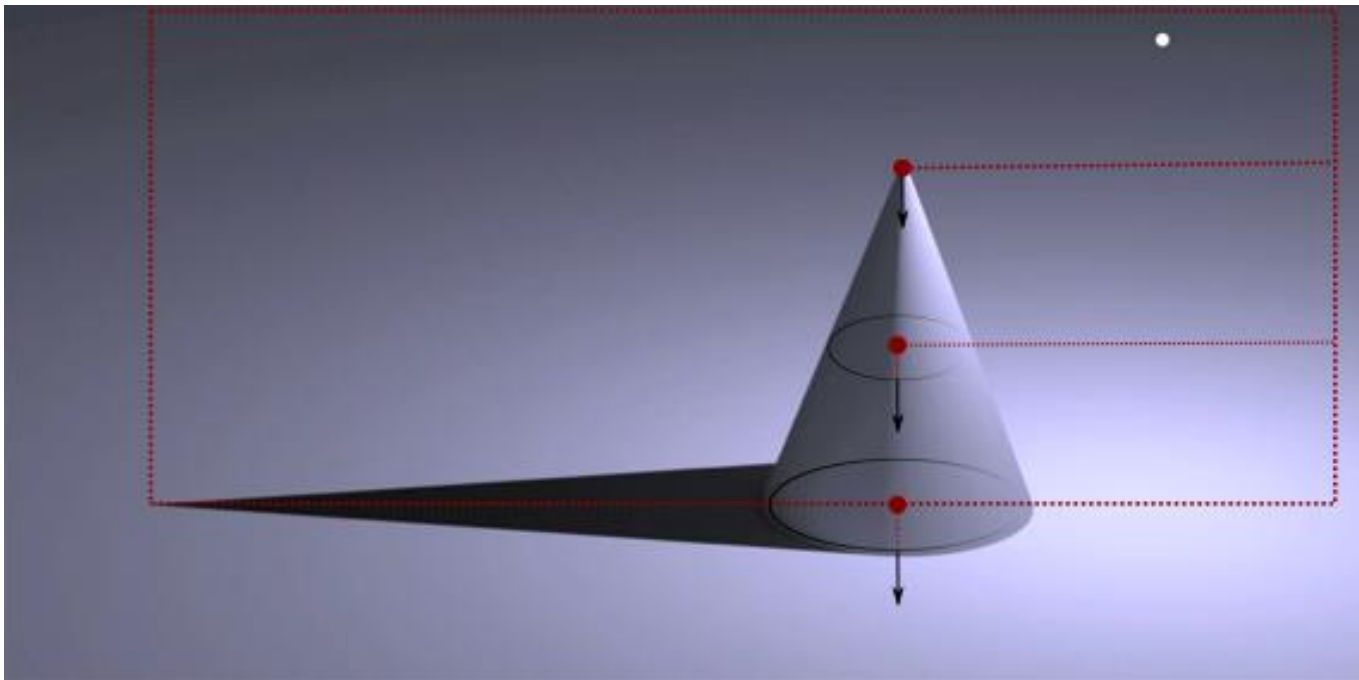


# Photo manipulation with inconsistent shadows

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Attached shadow

Easy case

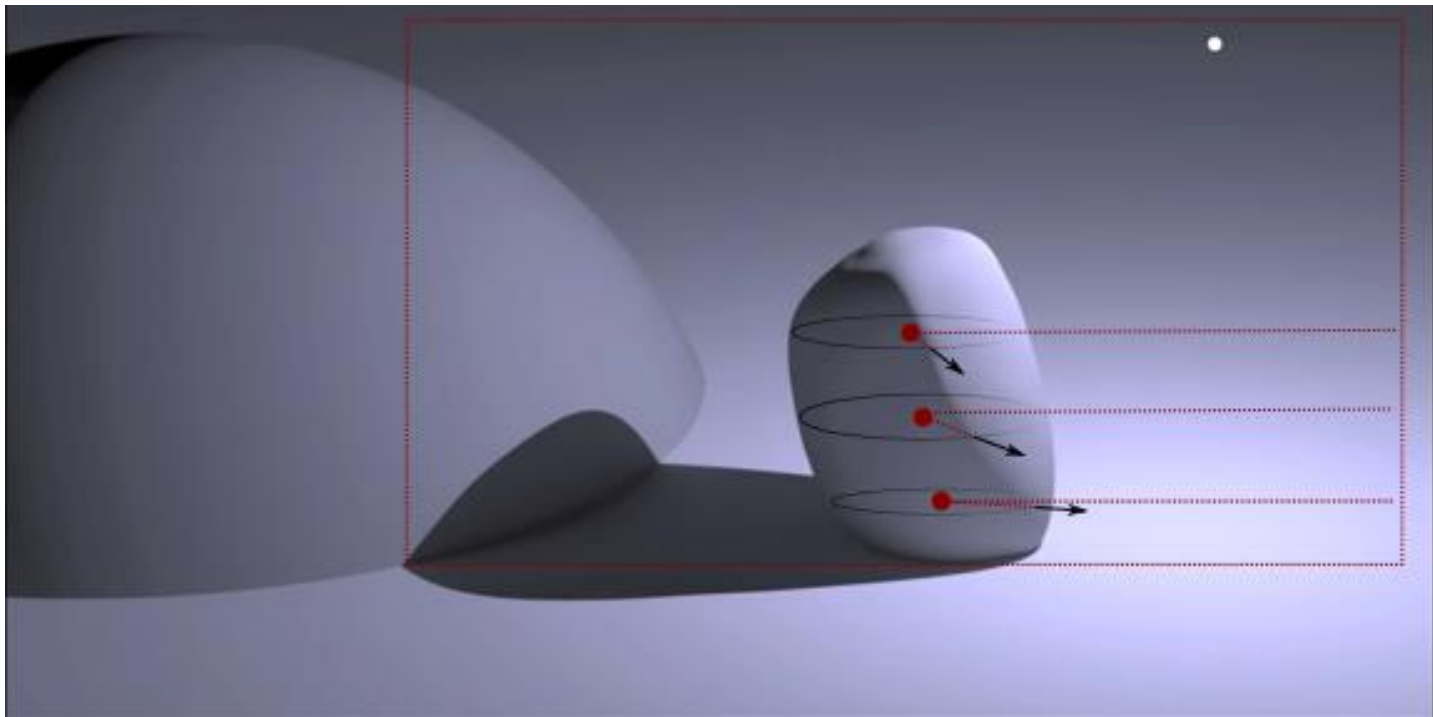


# Photo manipulation with inconsistent shadows

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Attached shadow

Difficult case



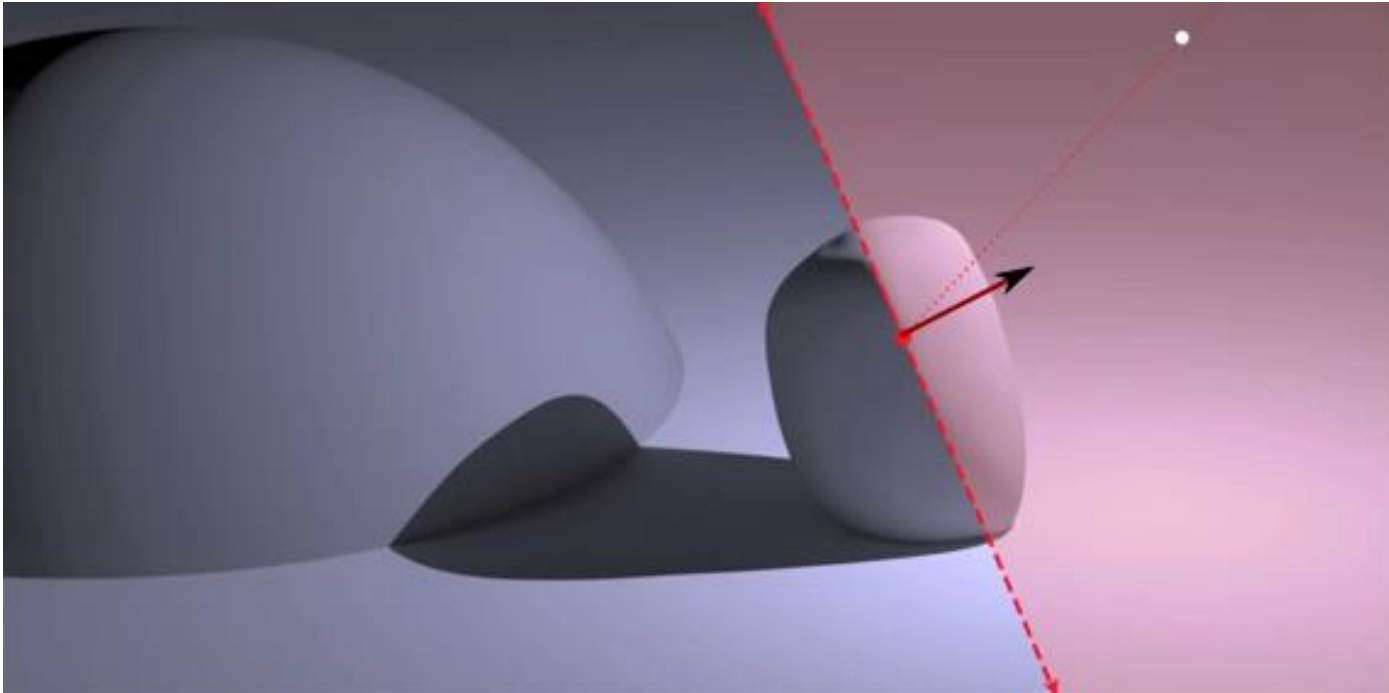
# Photo manipulation with inconsistent shadows

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## Attached shadow

Where is the light source?

- Solution: constrain the position to a region inside a half plane



# Photo manipulation with inconsistent shadows

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## Basic assumptions

In case of multi-objects scene:

- Mark every object
- The single light source is expected to be within the region formed by the intersection of all the constraints



# Photo manipulation with inconsistent shadows

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## Math background

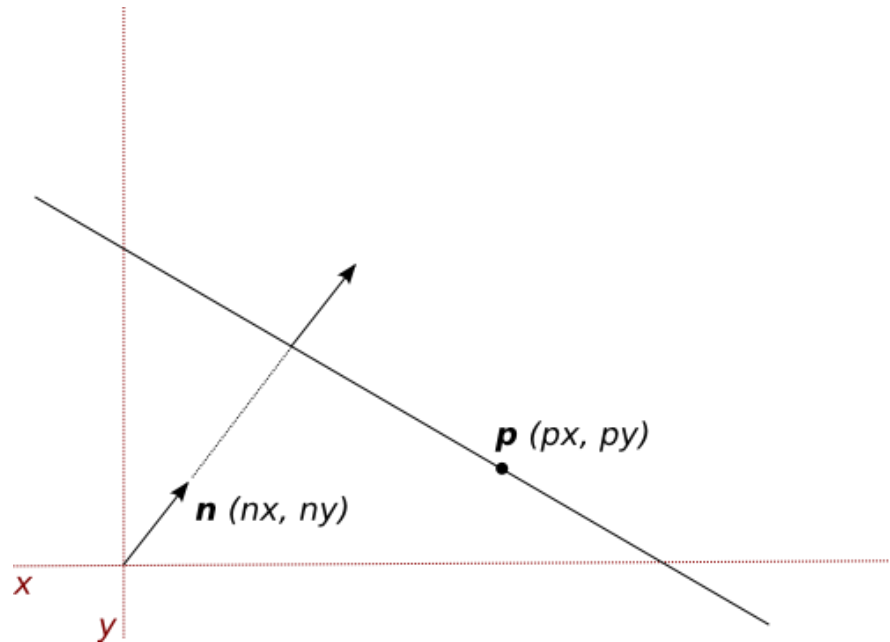
### Using analytic geometry

- Formula of a 2-D line passing through a known point  $p(px, py)$ , with a given normal  $n(nx, ny)$ :

$$nx \times x + ny \times y - (nx \times px + ny \times py) = 0 \therefore$$

$$\begin{pmatrix} nx \\ ny \end{pmatrix} \bullet \begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} nx \\ ny \end{pmatrix} \bullet \begin{pmatrix} px \\ py \end{pmatrix} = 0 \therefore$$

$$\mathbf{n} \bullet \mathbf{x} - \mathbf{n} \bullet \mathbf{p} = 0$$



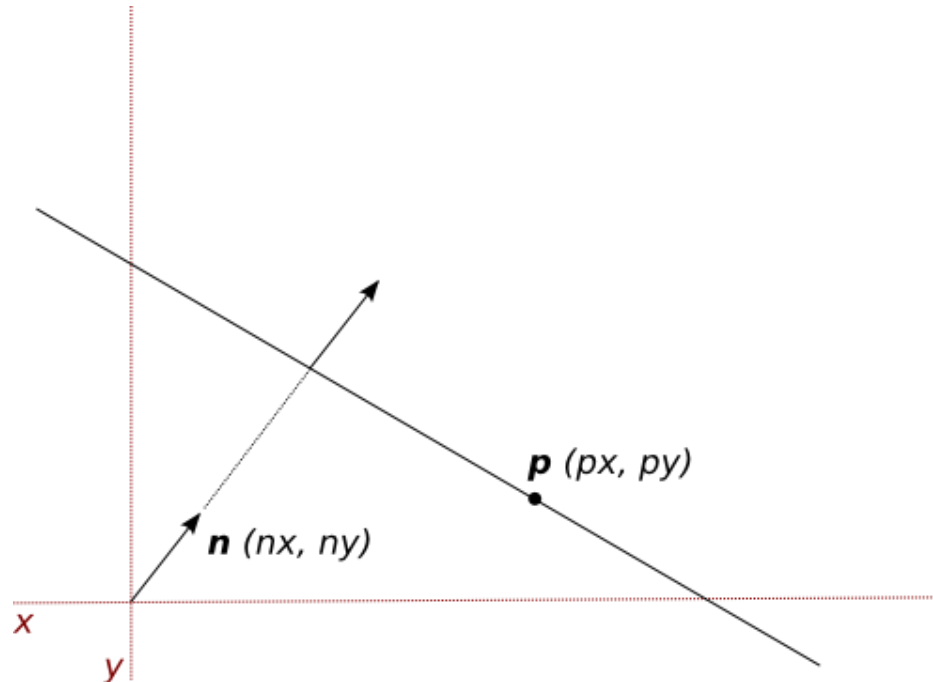
# Photo manipulation with inconsistent shadows

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## Math background

Using analytic geometry

- For the case of half-plane:  $\mathbf{n} \bullet \mathbf{x} - \mathbf{n} \bullet \mathbf{p} \geq 0$



# Photo manipulation with inconsistent shadows

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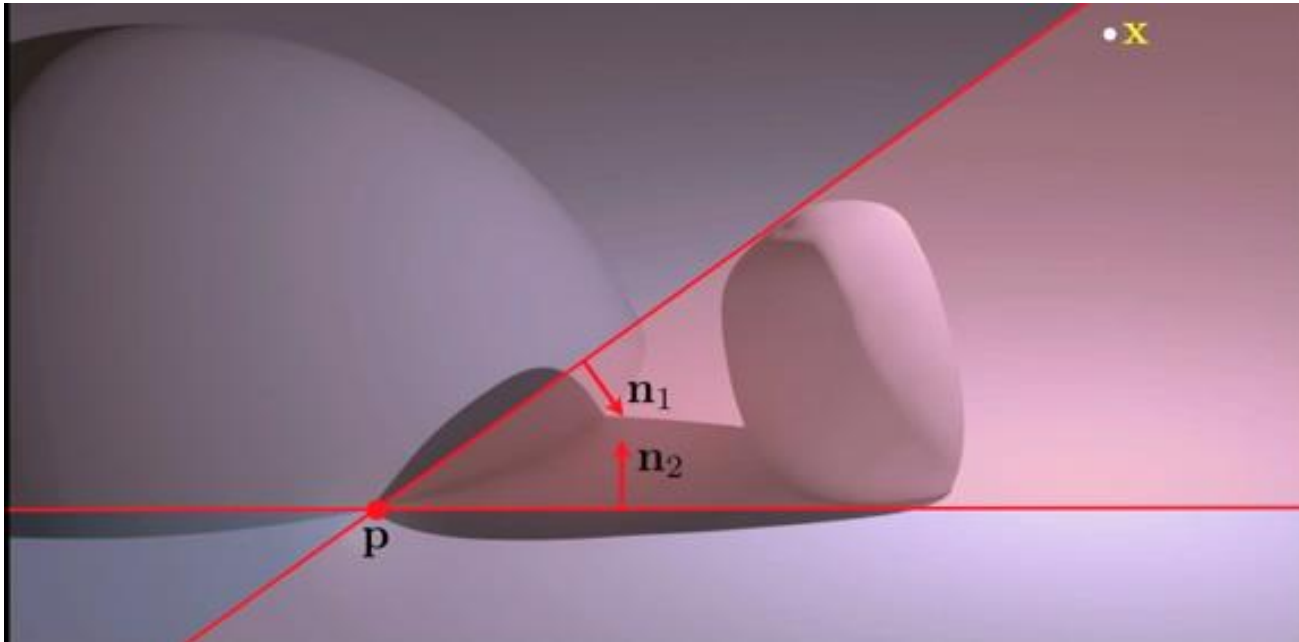
Math background

$$\mathbf{n}_1 \bullet \mathbf{x} - \mathbf{n}_1 \bullet \mathbf{p} \geq 0$$

Model applied to cast shadows:

*and*

$$\mathbf{n}_2 \bullet \mathbf{x} - \mathbf{n}_2 \bullet \mathbf{p} \geq 0$$

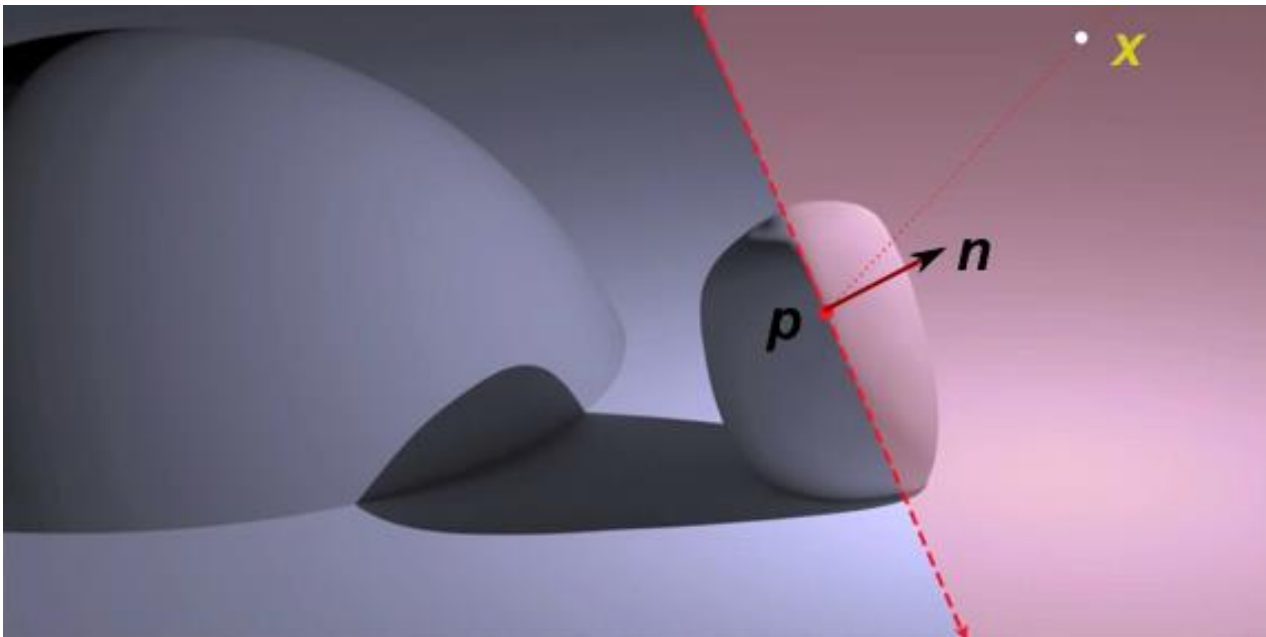


# Photo manipulation with inconsistent shadows

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## Math background

Model applied to attached shadows:  $\mathbf{n} \bullet \mathbf{x} - \mathbf{n} \bullet \mathbf{p} \geq 0$



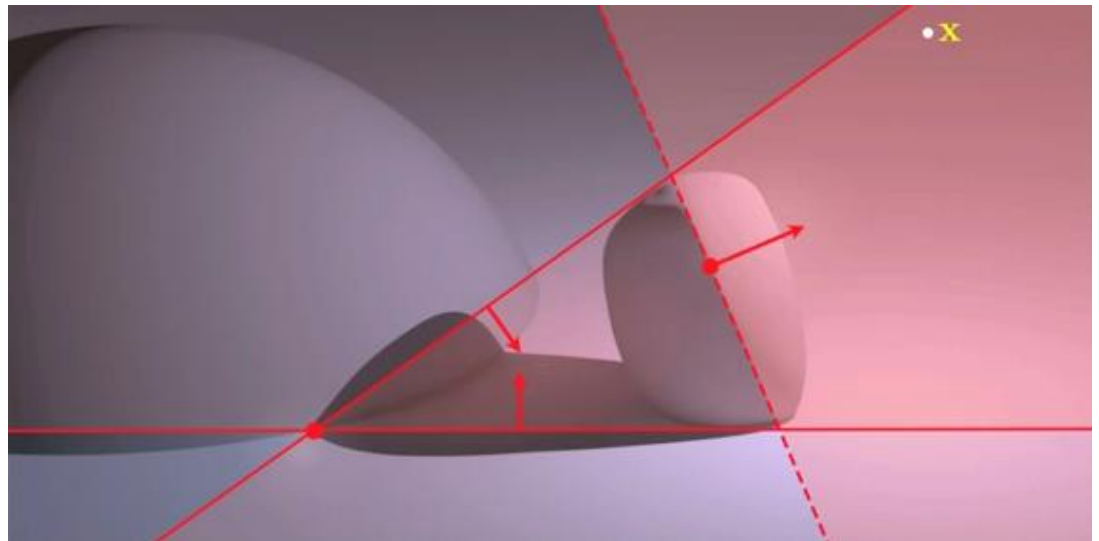
# Photo manipulation with inconsistent shadows

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## Math background

Combining models of cast and attached shadows together

$$\begin{pmatrix} \begin{pmatrix} nx_1 \\ ny_1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} \\ \begin{pmatrix} nx_2 \\ ny_2 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} \\ \begin{pmatrix} nx_3 \\ ny_3 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} \\ \vdots \\ \begin{pmatrix} nx_m \\ ny_m \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} \end{pmatrix} - \begin{pmatrix} \begin{pmatrix} nx_1 \\ ny_1 \end{pmatrix} \cdot \begin{pmatrix} px_1 \\ py_1 \end{pmatrix} \\ \begin{pmatrix} nx_2 \\ ny_2 \end{pmatrix} \cdot \begin{pmatrix} px_2 \\ py_2 \end{pmatrix} \\ \begin{pmatrix} nx_3 \\ ny_3 \end{pmatrix} \cdot \begin{pmatrix} px_3 \\ py_3 \end{pmatrix} \\ \vdots \\ \begin{pmatrix} nx_m \\ ny_m \end{pmatrix} \cdot \begin{pmatrix} px_m \\ py_m \end{pmatrix} \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$



# Photo manipulation with inconsistent shadows

## Math background

With following simplifications

$$\begin{pmatrix} \begin{pmatrix} nx_1 \\ ny_1 \end{pmatrix} \bullet \begin{pmatrix} x \\ y \end{pmatrix} \\ \begin{pmatrix} nx_2 \\ ny_2 \end{pmatrix} \bullet \begin{pmatrix} x \\ y \end{pmatrix} \\ \begin{pmatrix} nx_3 \\ ny_3 \end{pmatrix} \bullet \begin{pmatrix} x \\ y \end{pmatrix} \\ \vdots \\ \begin{pmatrix} nx_m \\ ny_m \end{pmatrix} \bullet \begin{pmatrix} x \\ y \end{pmatrix} \end{pmatrix} - \begin{pmatrix} \begin{pmatrix} nx_1 \\ ny_1 \end{pmatrix} \bullet \begin{pmatrix} px_1 \\ py_1 \end{pmatrix} \\ \begin{pmatrix} nx_2 \\ ny_2 \end{pmatrix} \bullet \begin{pmatrix} px_2 \\ py_2 \end{pmatrix} \\ \begin{pmatrix} nx_3 \\ ny_3 \end{pmatrix} \bullet \begin{pmatrix} px_3 \\ py_3 \end{pmatrix} \\ \vdots \\ \begin{pmatrix} nx_m \\ ny_m \end{pmatrix} \bullet \begin{pmatrix} px_m \\ py_m \end{pmatrix} \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \therefore$$

$$\begin{pmatrix} nx_1 & ny_1 \\ nx_2 & ny_2 \\ nx_3 & ny_3 \\ \vdots & \vdots \\ nx_m & ny_m \end{pmatrix} \times \begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} \begin{pmatrix} nx_1 \\ ny_1 \end{pmatrix} \bullet \begin{pmatrix} px_1 \\ py_1 \end{pmatrix} \\ \begin{pmatrix} nx_2 \\ ny_2 \end{pmatrix} \bullet \begin{pmatrix} px_2 \\ py_2 \end{pmatrix} \\ \begin{pmatrix} nx_3 \\ ny_3 \end{pmatrix} \bullet \begin{pmatrix} px_3 \\ py_3 \end{pmatrix} \\ \vdots \\ \begin{pmatrix} nx_m \\ ny_m \end{pmatrix} \bullet \begin{pmatrix} px_m \\ py_m \end{pmatrix} \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \therefore$$

$$\begin{pmatrix} \mathbf{n}_1^t \\ \mathbf{n}_2^t \\ \mathbf{n}_3^t \\ \vdots \\ \mathbf{n}_m^t \end{pmatrix} \times \begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} \mathbf{n}_1 \bullet \mathbf{p}_1 \\ \mathbf{n}_2 \bullet \mathbf{p}_2 \\ \mathbf{n}_3 \bullet \mathbf{p}_3 \\ \vdots \\ \mathbf{n}_m \bullet \mathbf{p}_m \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \therefore$$

$$\mathbf{N} \times \mathbf{x} - \mathbf{P} \geq \mathbf{0}$$

# Photo manipulation with inconsistent shadows

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Math background

Problem solution

- solution to this system should always exist
- in case of inconsistencies slack variables are added

$$\begin{pmatrix} s_1 \\ s_2 \\ s_3 \\ \vdots \\ s_m \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad \therefore \mathbf{s} \geq \mathbf{0}$$

$$\begin{pmatrix} \mathbf{n}_1^t \\ \mathbf{n}_2^t \\ \mathbf{n}_3^t \\ \vdots \\ \mathbf{n}_m^t \end{pmatrix} \times \begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} \mathbf{n}_1 \bullet \mathbf{p}_1 \\ \mathbf{n}_2 \bullet \mathbf{p}_2 \\ \mathbf{n}_3 \bullet \mathbf{p}_3 \\ \vdots \\ \mathbf{n}_m \bullet \mathbf{p}_m \end{pmatrix} + \begin{pmatrix} s_1 \\ s_2 \\ s_3 \\ \vdots \\ s_m \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad \therefore \mathbf{N} \times \mathbf{x} - \mathbf{P} + \mathbf{s} \geq \mathbf{0} \quad \therefore \mathbf{N} \times \mathbf{x} - \mathbf{P} \geq -\mathbf{s}$$



# Photo manipulation with inconsistent shadows

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## Math background

After solving the linear programming problem:

- if slack variables for optimal solution are all ZERO: there is a single light source position that satisfies all the constraints
- else, one or more shadows is inconsistent with the single light source assumption

## Caution:

Due to the adopted projective model, there is an inherent sign ambiguity when specifying the shadow constraints

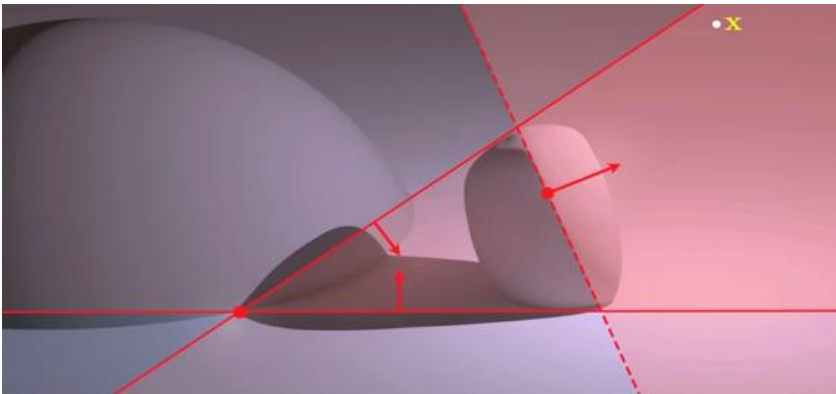
- If the light source in the real 3-D world is behind the camera:
  - Its position is inversely projected onto the image plane
  - All constraint normals should be negated

# Photo manipulation with inconsistent shadows

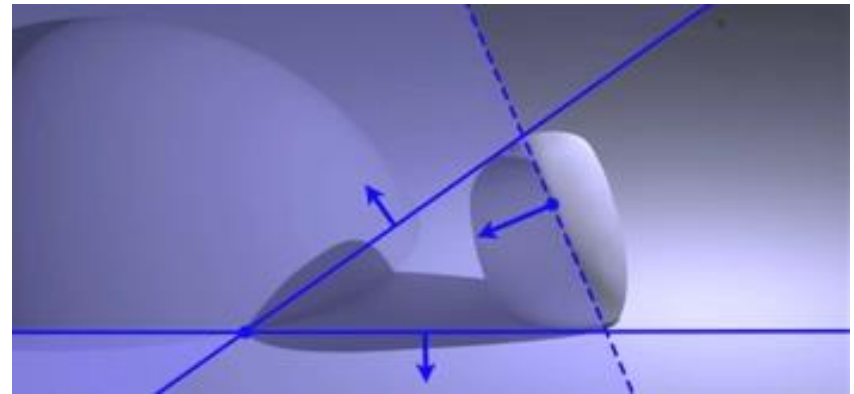
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## Math background

In order to not worry about the position of the light source in the real world, the authors solve two kinds of linear programs:



$$\mathbf{N} \times \mathbf{x} - \mathbf{P} \geq -s$$



$$-\mathbf{N} \times \mathbf{x} - \mathbf{P} \geq -s$$

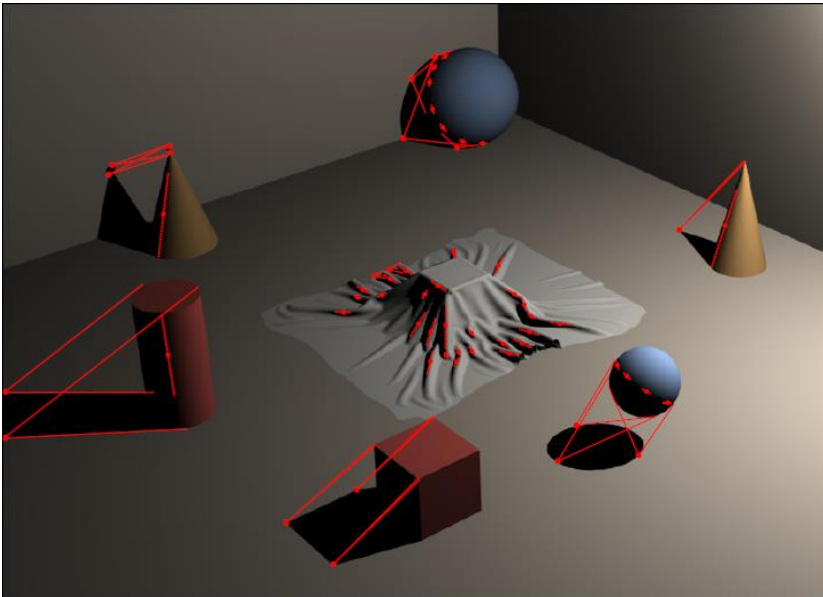
# Photo manipulation with inconsistent shadows

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## Method application

Method can be applied to:

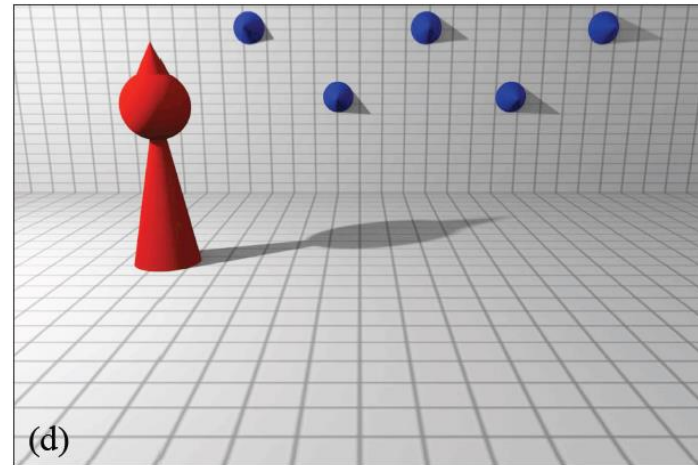
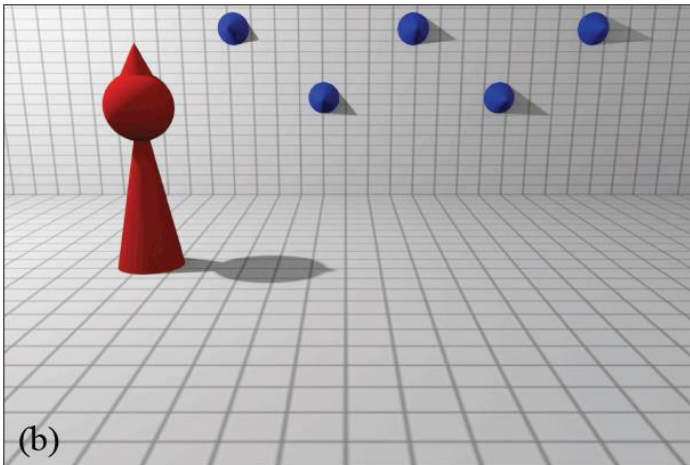
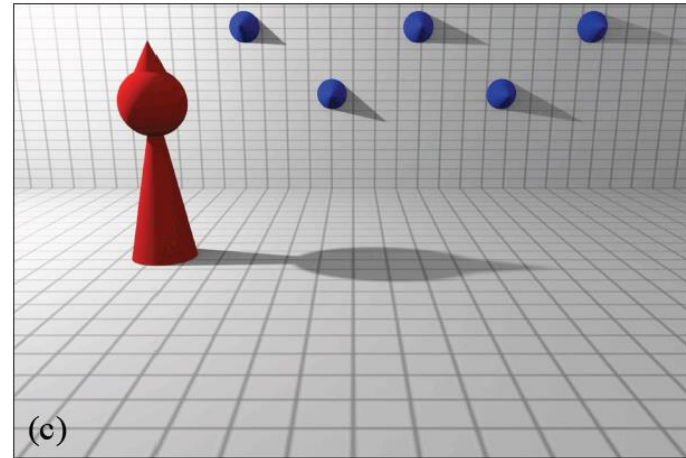
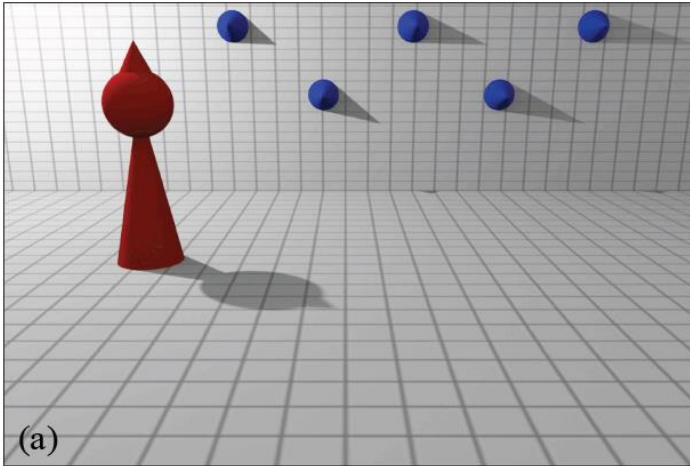
- Simulations and real-world examples
- Authentic and forgeries



# Photo manipulation with inconsistent shadows

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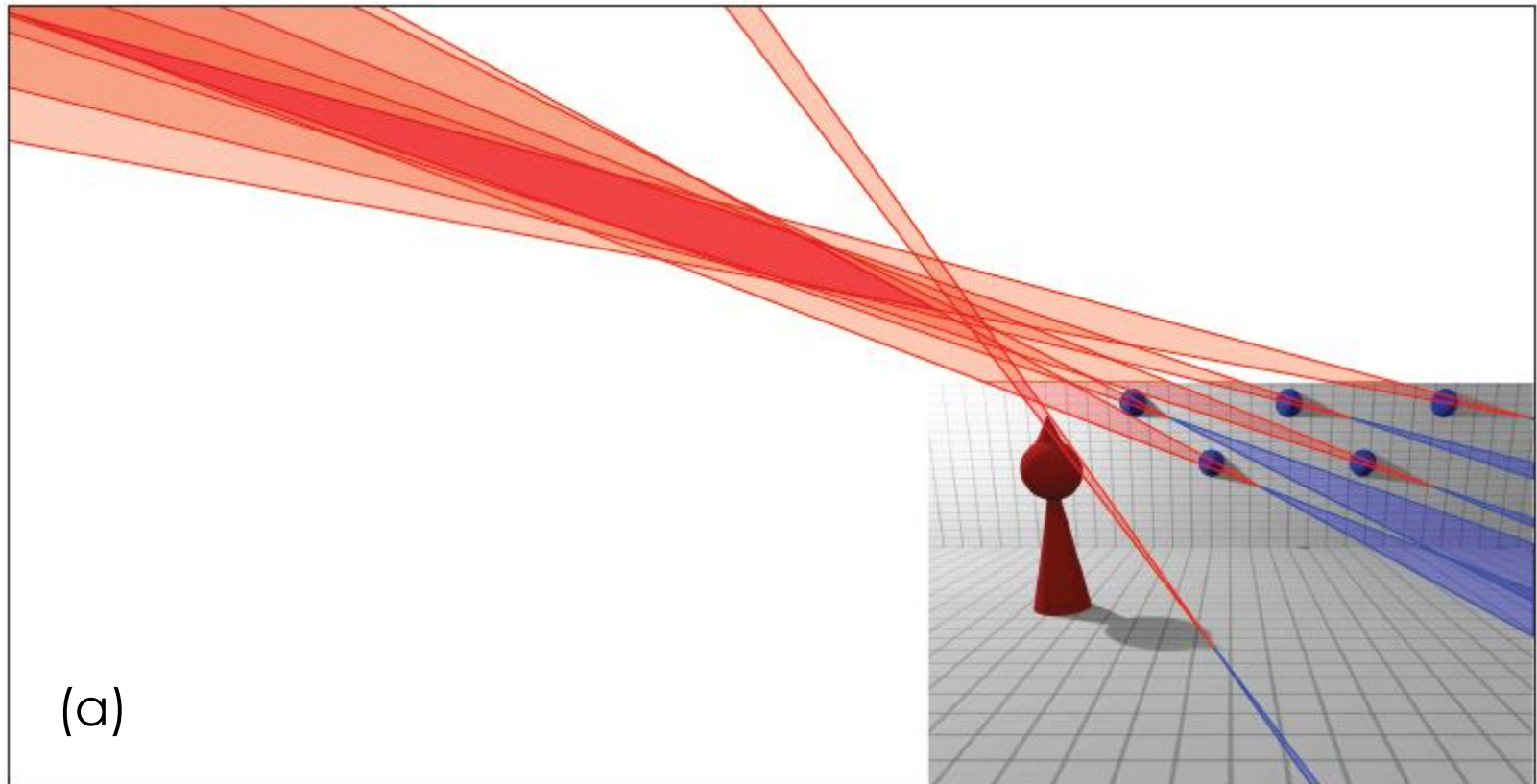
## Method application (test case)



# Photo manipulation with inconsistent shadows

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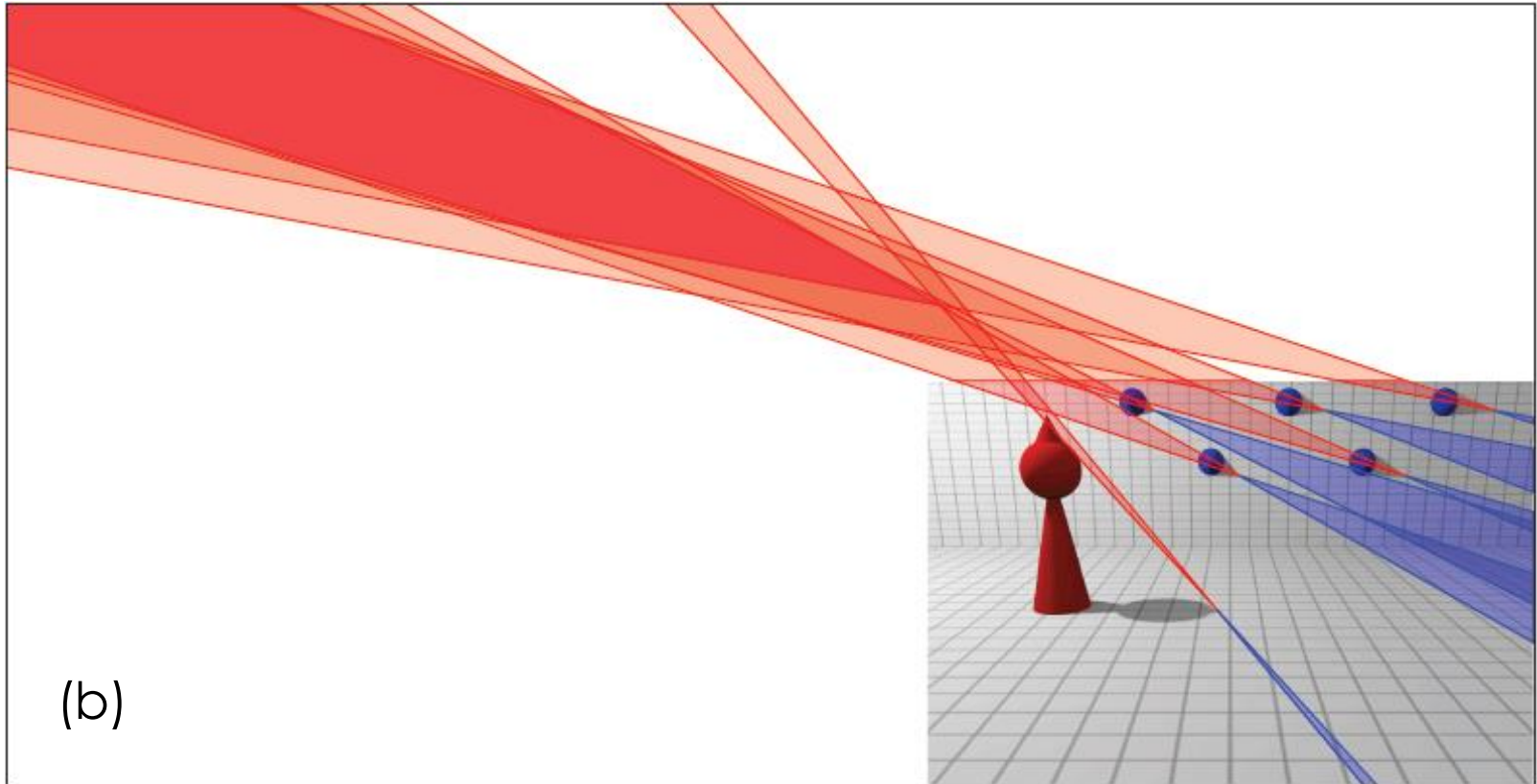
## Method application (test case)



# Photo manipulation with inconsistent shadows

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Method application (test case)

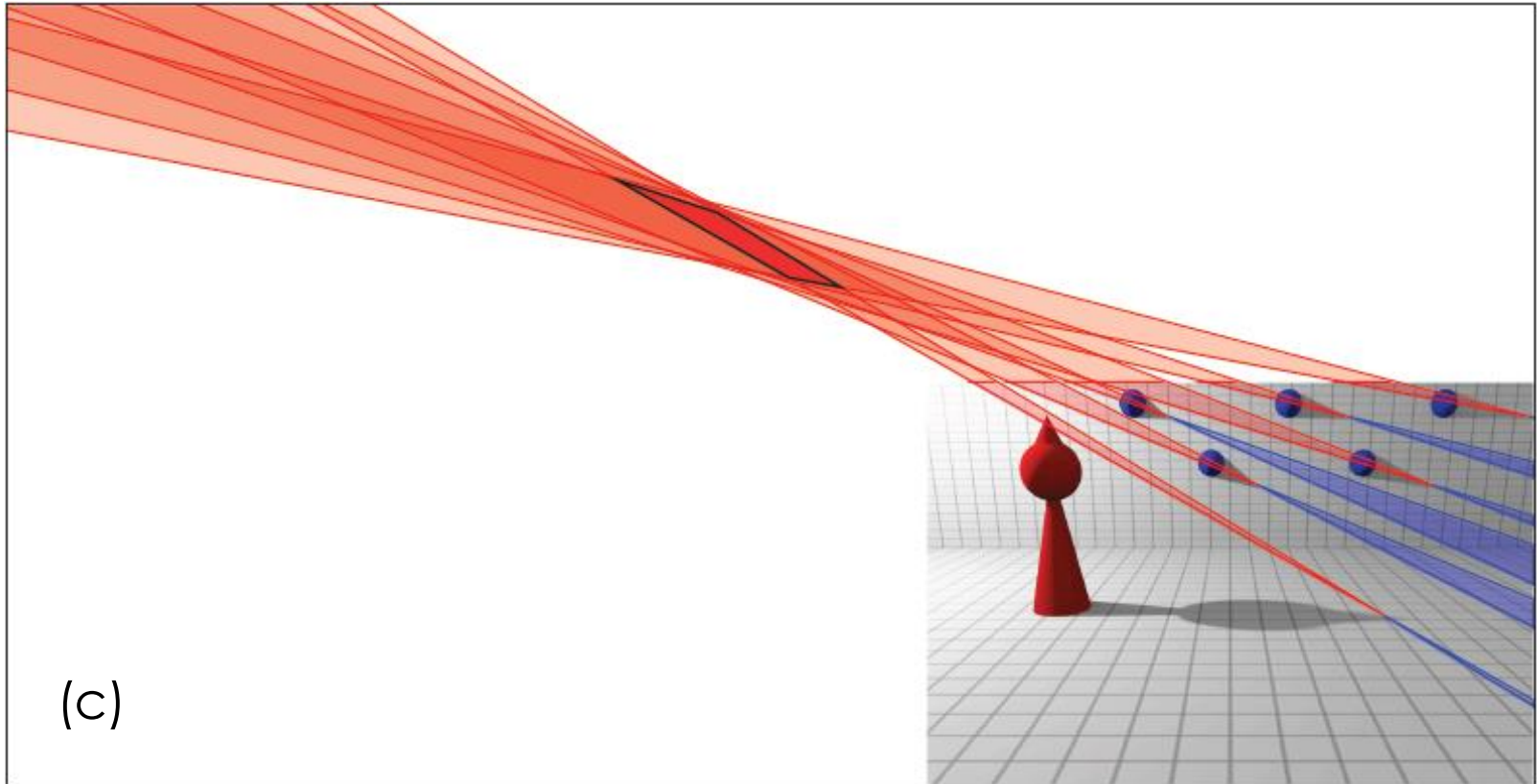




# Photo manipulation with inconsistent shadows

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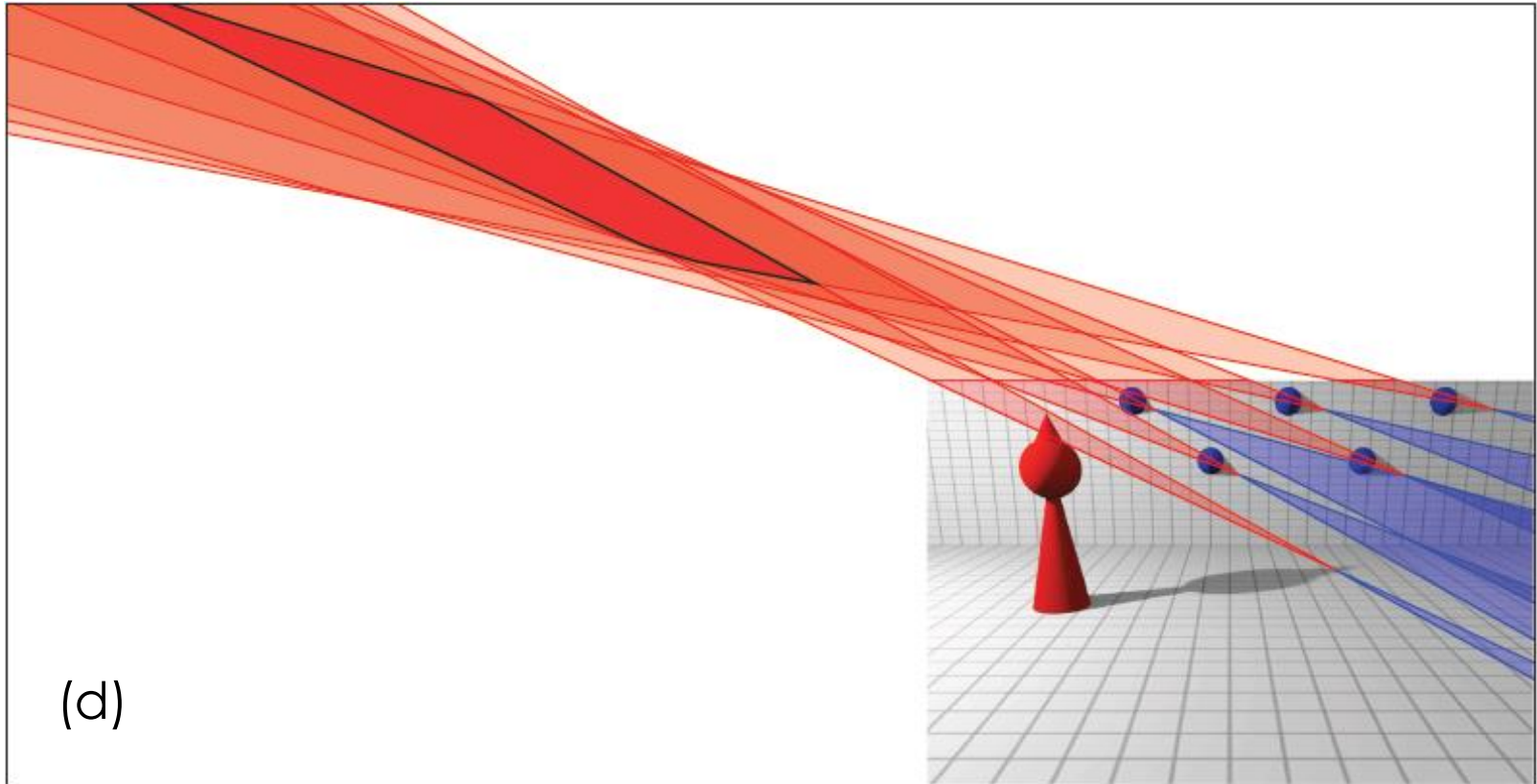
Method application (test case)



# Photo manipulation with inconsistent shadows

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Method application (test case)

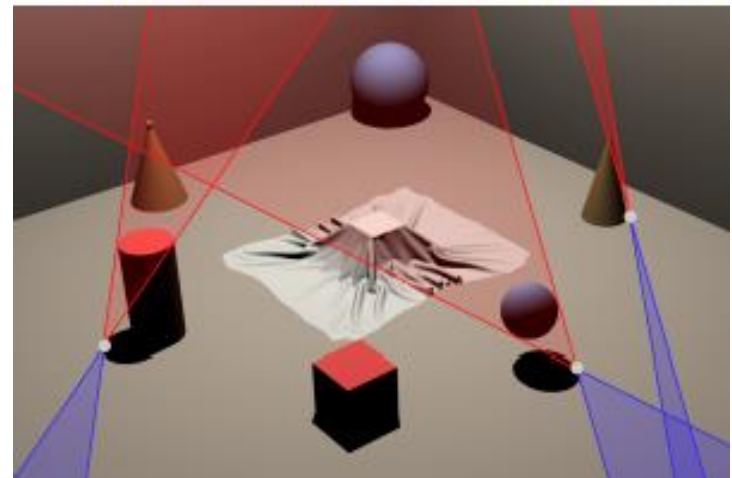
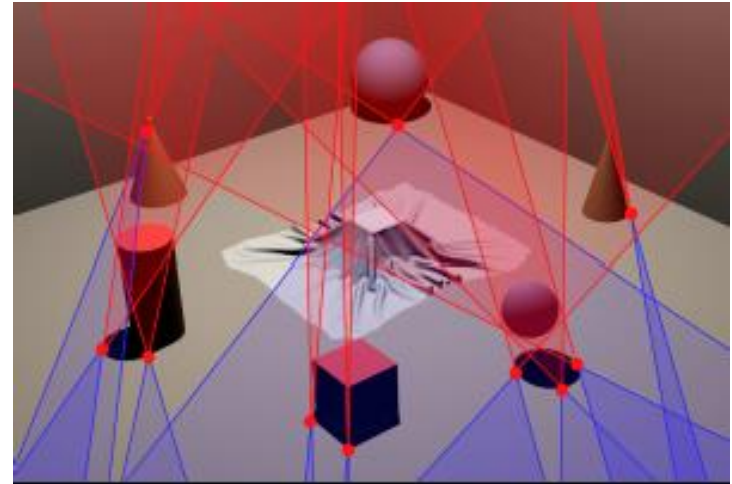
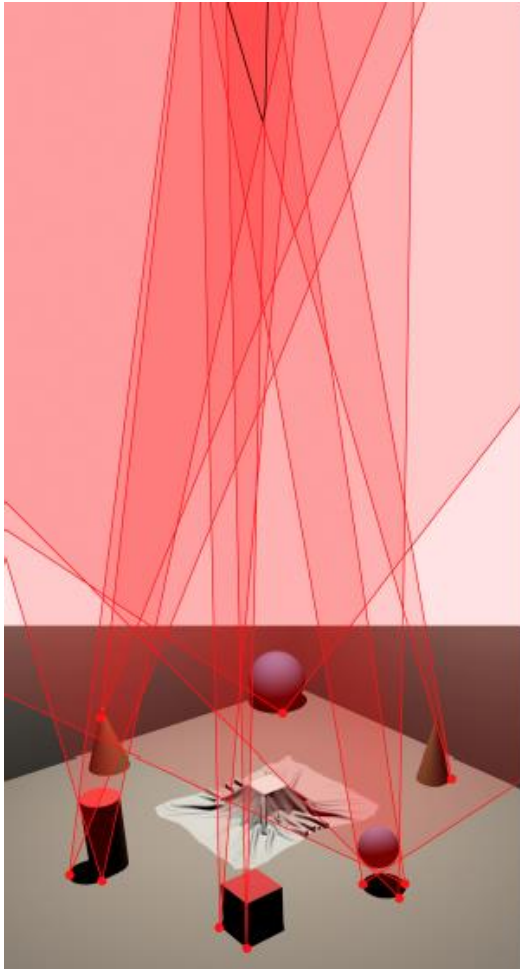




# Photo manipulation with inconsistent shadows

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## Method application (consistent shadows)



# Photo manipulation with inconsistent shadows

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## Method application (inconsistent shadows)



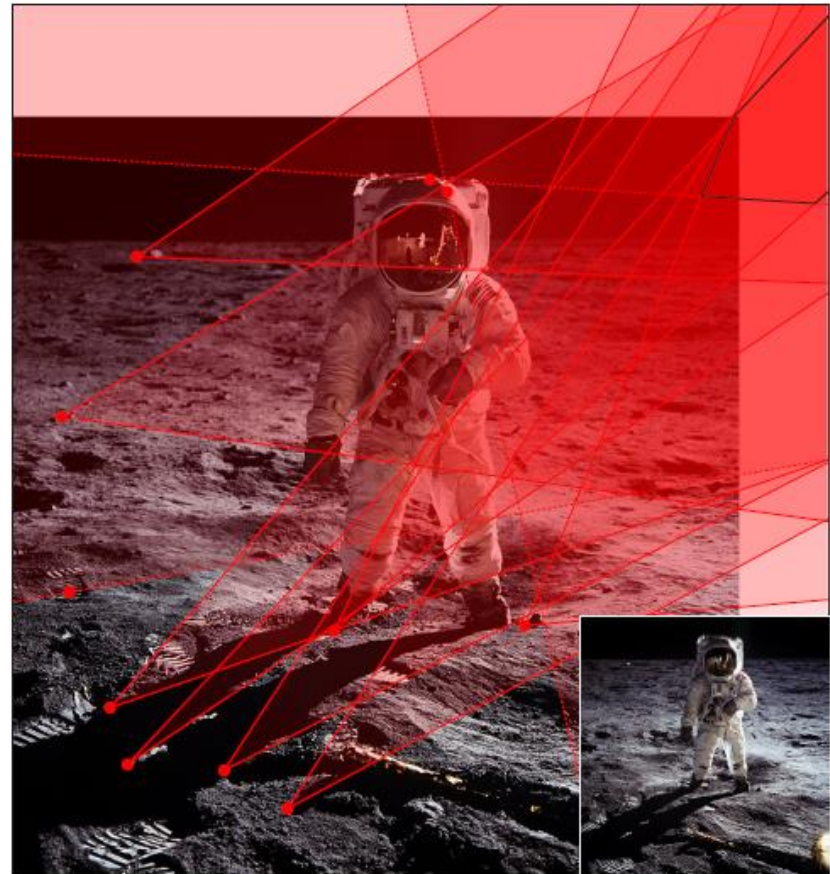
*Shadows on the angel are inconsistent with the cloth and the angel's cast shadow*

# Photo manipulation with inconsistent shadows

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## Method application (Moon photo case)

- Region outlined in black identifies locations that satisfy applied constraints
- By this measure, the lighting in this image is consistent





# Photo manipulation with inconsistent shadows

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## Method application (real-life case)

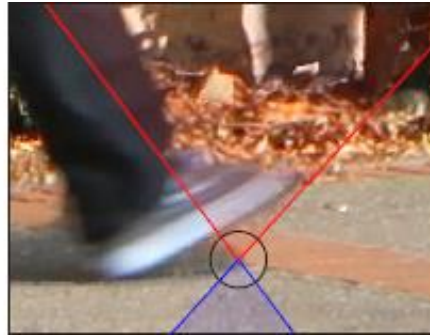


# Photo manipulation with inconsistent shadows

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## Method application (real-life case)

- Magnified view of the selected shadows (only a minimal set of violating constraints)





# Photo manipulation with inconsistent shadows

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## Method application (real-life case)



# Photo manipulation with inconsistent shadows

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## Conclusions

- Broad range of ambiguous cast and attached shadows
  - Combination of many constraints can be effective
- Easy to identify inconsistent constraints.
- Computationally efficient solution (standard linear programming problem)
- Works only for a single dominant light source
  - Preclude analysis of scenes with multiple point lights or diffuse area lighting
  - Includes common situation of outdoor scenes or indoor scenes photographed with a flash



# Photo manipulation with inconsistent shadows

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## Conclusions

- No assumptions about the scene geometry or photometry
  - Only linear perspective projection and a single dominant point light source
- Lens distortion may be an issue
- Depends on the user
  - Poor selection of constraints could lead to a failure in detecting a manipulation
- Counter forensics
  - Using this technique to ensure that all shadows are consistent
  - Combine with other techniques for estimating lighting from a single image

# Photo manipulation with inconsistent shadows

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